# Licensing Resources and Documents

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# Windows Server 2025 licensing guidance

## Summary

This guide can help Microsoft Commercial Licensing customers (Volume Licensing) understand how Windows Server 2025 is licensed through Microsoft Commercial Licensing programs. This guide is for informational purposes and does not supersede or replace any documentation covering Windows Server 2025 licensing.

Specific license terms for Windows Server 2025 are defined in the Microsoft Commercial Licensing Product Terms, the Microsoft Commercial Licensing agreement under which it was acquired, and/or its original equipment manufacturer (OEM) or Retail Software License Terms. This licensing guide is not a legal use rights document. Program specifications and business rules are subject to change.

For complete details and information on licensing, refer to the Product Terms for Commercial Licensing use rights, or the Microsoft License Terms for OEM and retail. Additional information about Windows Server 2025, including the latest version of this guide, is available on the Microsoft Licensing Resources and Documents site, Windows Server page.

## Product overview

Windows Server 2025 is the cloud-ready operating system customers can use to securely run their workloads, enable new hybrid cloud scenarios, and modernize their applications to meet evolving business requirements. It brings customers advanced security and resiliency, hybrid cloud agility, and a future-ready, high-performance infrastructure.

**Advanced multi-layered security** Customers can take advantage of multi-layer security with Secured-core server and secured connectivity. Secured-core server means our hardware partners have provided hardware, firmware, and drivers to help customers harden the security of their critical systems. It allows IT and SecOps teams to apply comprehensive security broadly in their environment with Secured-core server's advanced protection and preventive defense across hardware, firmware, and virtualization layers.

**Hybrid capabilities with Azure** Customers choosing a hybrid and multicloud approach to digitally transform their businesses can now take advantage of cloud services with on-premises Windows Server 2025 by connecting with Azure Arc, and Hotpatching is available for on-premises Windows Server 2025

when connected to Azure Arc. Additionally, Windows Server 2025 customers can take advantage of the File Server enhancements. SMB Compression improves application file transfer by compressing data while in transit over a network. Finally, Windows Admin Center, a tool loved by admins, brings modern server management experience with a new event viewer and gateway proxy support for Azure connected scenarios.

**Flexible application platform** Customers who upgrade to Windows Server 2025 can take advantage of scalability improvements such as support for 48TB of memory and 2,048 logical cores running on 64 physical sockets for those demanding Tier1 applications. In this release, customers can also take advantage of advancements to Windows containers. For example, Windows Server 2025 improves application compatibility of Windows containers, includes HostProcess containers for node configuration, supports IPv6 and dual-stack, and enables consistent network policy implementation with Calico.

# Edition overview and comparison

The Windows Server 2025 product is streamlined and simple, making it easy for customers to choose the edition that is right for their needs. Choose from three primary editions of Windows Server, based on organization size as well as virtualization and datacenter requirements:

**Datacenter edition** is ideal for highly virtualized and software-defined datacenter environments.

**Standard edition** is ideal for customers with low density or non-virtualized environments.

**Essentials edition** is a cloud-connected first server, ideal for small businesses with up to 25 users and 50 devices. Windows Server 2025 Essentials edition is available to purchase from OEMs only.

A Windows Server 2025 Standard and Datacenter edition comparison can be viewed here: Windows Server 2025 Licensing & Pricing | Microsoft

# Server licensing and feature overview

Windows Server 2025 Datacenter and Windows Server 2025 Standard are licensed under a core-based license model. For both Datacenter and Standard, the number of core licenses required depends on whether a customer is licensing based on the physical cores on the server or by virtual machine. The option to license Windows Server by virtual machine was added in October 2022 and is available to customers with subscription licenses or licenses with active Software Assurance only.

- When licensing based on the physical cores on the server, the number of core licenses required equals the number of physical cores on the licensed server, subject to a minimum of 8 core licenses per physical processor and a minimum of 16 core licenses per server.
- When licensing by virtual machine, the number of core licenses required equals the number of virtual cores in the virtual operating system environment (i.e., virtual machine), subject to a minimum of 8 core licenses per virtual machine.

Core licenses are sold in 2-packs and 16-packs. For complete details and information on licensing, refer to the Product Terms. To learn more about core licensing see the Introduction to Microsoft Core Licensing licensing brief.

| Attribute | Datacenter | Standard | Essentials |
|---|---|---|---|
| Licensing model | Per Core/CAL [1] | Per Core/CAL [1] | OEM only [2] |
| License type | Core license | Core license | Server license |
| OSEs/Hyper-V containers | Unlimited/One [3] | Two [4]/One [3] | One [5] |
| Windows Server containers | Unlimited | Unlimited | |

**Notes:**

[1] When licensing based on physical cores on the server, all physical cores on the server must be licensed, subject to a minimum of 8 core licenses per physical processor and a minimum of 16 core licenses per server. When licensing by virtual machine, the number of core licenses required equals the number of virtual cores in the virtual operating system environment, subject to a minimum of 8 core licenses per virtual machine and 16 core licenses per customer. Licensing by virtual machine requires subscription licenses or licenses with active Software Assurance.

[2] Windows Server Essentials edition server is for small businesses with up to 25 users and 50 devices. A server license covers up to 10 cores and 1 VM on single-socket servers. Windows Server Essentials is available through our OEM Server Hardware partners only.

[3] When licensed by virtual machine, both Windows Server Standard and Windows Server Datacenter use of Windows Server in one virtual OSE on the licensed server. Licensing by virtual machine requires subscription licenses or licenses with active Software Assurance.

[4] When licensed based on the physical cores on the server, Windows Server Standard permits use of the server software in the physical OSE on the licensed server (in addition to two virtual OSEs), if the physical OSE is used solely to host and manage the virtual OSEs

[5] Windows Server Essentials edition permits use of one running instance of the server software in the physical OSE on the licensed server (in addition to one virtual OSE), if the physical OSE is used solely to host and manage the virtual OSE.

# Virtualization Rights

## Datacenter edition

***Licensing based on physical cores***

As shown in the table above, when licensed based on the physical cores on a server, Windows Server Datacenter provides rights to use any number of operating system environments (physical or virtual OSEs) or Windows Server Containers with Hyper-V isolation as well as any number of Windows Server containers without Hyper-V isolation on that server.

***Licensing by virtual machine***

When licensed by virtual machine, Windows Server Datacenter provides rights to use the software in either one virtual OSE, one Windows Server container with Hyper-V isolation or any number of Windows Server containers without Hyper-V isolation in one virtual OSE.

## Standard edition

***Licensing based on physical cores***

When licensed based on the physical cores on a server, Windows Server Standard provides rights to use two operating system environments (physical or virtual OSEs) or Windows Server Containers with Hyper-V isolation and unlimited Windows Server Containers without Hyper-V isolation on that server. When licensed based on physical cores, Standard edition also permits use of the server software in the

physical OSE on the licensed server (in addition to two virtual OSEs), if the physical OSE is used solely to host and manage the virtual OSEs.

***Licensing by virtual machine***

As with Datacenter, when licensed by virtual machine, Windows Server Standard provides rights to use the software in either one virtual OSE, one Windows Server Container with Hyper-V isolation or any number of Windows Server Containers without Hyper-V isolation in one virtual OSE.

For example, a 2-processor server with 8 cores per processor requires 16 core licenses (for example, one 16-pack of core licenses or eight 2-packs of core licenses) and gives rights to two OSEs (physical or virtual) or two Windows Server Containers with Hyper-V isolation. In the case of this example, for each additional two OSEs or two Windows Server Containers with Hyper-V isolation the customer wishes to use, an additional 16 core licenses must be assigned to the server. Alternatively, the customer could license additional OSEs or containers by virtual machine.

### Essentials edition

Each server license provides rights to use one running instance of the server software in either a physical OSE or virtual OSE on the licensed server.

# Determining the required number of core licenses

Server software licensing for Windows Server 2025 Standard and Datacenter requires core licenses. Customers may choose between licensing based on physical cores or licensing by virtual machine. Licensing by virtual machine is available only under subscription licenses or licenses with active Software Assurance. For each server licensed based on physical cores, the number of core licenses required equals the number of physical cores on the licensed server, subject to a minimum of 8 core licenses per physical processor and a minimum of 16 core licenses per server. For each server licensed by virtual machine, the number of core licenses required equals the number of virtual cores in a virtual operating system environment, subject to a minimum of 8 core licenses per virtual machine. Core licenses are sold in 2-packs and 16-packs. For complete details on server licensing, refer to the Product Terms.

## Minimum requirements for Standard and Datacenter editions (licensed based on physical cores)

The table below provides examples for various server configurations, and the minimum number of core licenses required.

**Standard:** When all cores on the server are licensed (subject to a minimum of 8 core licenses per physical processor and a minimum of 16 core licenses per server), Standard has rights to use two OSEs or two Windows Server containers with Hyper-V isolation and unlimited Windows Server containers without Hyper-V isolation.

**Datacenter:** When all cores on the server are licensed (subject to a minimum of 8 core licenses per physical processor and a minimum of 16 core licenses per server), Datacenter edition has rights to use unlimited OSEs, Windows Server containers with Hyper-V isolation and Windows Server containers without Hyper-V isolation.

| Server licensing | 1-processor server | 1-processor server | 2-processor server | 2-processor server | 4-processor server | 4-processor server |
|---|---|---|---|---|---|---|
| Windows Server Standard & Datacenter | Required # cores licenses [1] | Required # 2-pack SKUs [2] | Required # cores licenses [1] | Required # 2-pack SKUs [2] | Required # cores licenses [1] | Required # 2-pack SKUs [2] |
| 2 cores per processor | 16 | 8 | 16 | 8 | 32 | 16 |
| 4 cores per processor | 16 | 8 | 16 | 8 | 32 | 16 |
| 6 cores per processor | 16 | 8 | 16 | 8 | 32 | 16 |
| 8 cores per processor | 16 | 8 | 16 | 8 | 32 | 16 |
| 10 cores per processor | 16 | 8 | 20 | 10 | 40 | 20 |

**Notes:**

[1] Core licenses are sold in 2-packs as well as optional 16-packs (for convenience).

[2] Eight 2-packs and one 16-pack are priced the same and provide equivalent use rights.

# Licensing requirements of additional OSEs for Standard edition

When licensed based on physical cores, Windows Server Standard has rights to use two operating system environments (OSEs) or two Windows Server containers with Hyper-V isolation and unlimited Windows Server containers without Hyper-V isolation (licenses equal to the physical cores on the server are assigned (subject to a minimum of 8 core licenses per physical processor and a minimum of 16 core licenses per server). Once a server is licensed, customers may wish to license the server for additional OSEs or Hyper-V containers. This practice is often referred to as “stacking” and is allowed with Standard edition.

The table below provides examples of “stacking” scenarios for various server configurations, the minimum number of licenses required, and the resulting number of OSEs or Hyper-V containers provided. As a rule, for each additional set of two OSEs or two Hyper-V containers the customer wishes to use, the server must be relicensed for the same number of core licenses. Note that Datacenter edition has rights to unlimited virtualization so “stacking” therefore is not required. Also, as an alternative to fully relicensing based on physical cores, customers with subscription licenses or licenses with active Software Assurance can license additional OSEs by virtual machine.

| ”Stacking” Standard | 1-proc server with 16 cores | 1-proc server with 16 cores | 2-proc server with 16 cores | 2-proc server with 16 cores | 4-proc server with 32 cores | 4-proc server with 32 cores |
|---|---|---|---|---|---|---|
| OSEs or | Required # | Required # | Required # | Required # | Required # | Required # |

| "Stacking" Standard | 1-proc server with 16 cores | 1-proc server with 16 cores | 2-proc server with 16 cores | 2-proc server with 16 cores | 4-proc server with 32 cores | 4-proc server with 32 cores |
|---|---|---|---|---|---|---|
| Hyper-V containers | cores licenses [1] | 2-pack SKUs [2] | cores licenses [1] | 2-pack SKUs [2] | cores licenses [1] | 2-pack SKUs [2] |
| 2 per server | 16 | 8 | 16 | 8 | 32 | 16 |
| 4 per server | 32 | 16 | 32 | 16 | 64 | 32 |
| 6 per server | 48 | 24 | 48 | 24 | 96 | 48 |
| 8 per server | 64 | 32 | 64 | 32 | 128 | 64 |
| 10 per server | 80 | 40 | 80 | 40 | 160 | 80 |

**Notes:**

[1] Core licenses are sold in 2-packs as well as optional 16-packs (for convenience).

[2] Eight 2-packs and one 16-pack are priced the same and provide equivalent use rights. However, point counts may vary, so MPSA and Open Volume customers should assess which pack sizes work best for them.

# Server access licensing overview

Whether Windows Server is licensed based on physical cores or by virtual machine, server software access requires access licenses. Access licenses include Client Access Licenses (CALs) and External Connectors. Both CALs and External Connectors have Base and Additive licenses. Generally, CALs, whether Base or Additive, can be User or Device. For complete details on server access licensing, refer to the Product Terms.

## Structure and application of access licenses

| Product | Class | Category | Type |
|---|---|---|---|
| Windows Server | Client Access License | Base | Per user or device |
| Windows Server Remote Desktop Services | Client Access License | Additive | Per user or device |
| Windows Server Active Directory Rights Management Services | Client Access License | Additive | Per user or device |
| Microsoft Identity Manager User | Client Access License | Additive | Per user |
| Windows Server | External Connector | Base | Per server [1] |
| Windows Server Remote Desktop Services | External Connector | Additive | Per server [1] |
| Windows Server Active Directory Rights Management Services | External Connector | Additive | Per server [1] |

| Product | Class | Category | Type |
|---|---|---|---|
| Microsoft Identity Manager User | External Connector | Additive | Per server [1] |

**Note:**

[1] Each physical server accessed requires an External Connector.

CALs are required for users that are either the customer's or its affiliates' employees, or its or its affiliates' onsite contractors or onsite agents. CALs or, alternatively, External Connectors are required for users that are not either the customer's or its affiliates' employees, or its or its affiliates' onsite contractors or onsite agents.

**CALs:** Each user or device that accesses licensed servers requires a Windows Server CAL (Base CAL). Each user or device that accesses advanced functionality on the licensed server requires the CAL associated with that functionality (Additive CAL). Additive CALs must be licensed in addition to the corresponding Base CALs for access to advanced functionality.

**External Connectors:** External access to licensed servers requires CALs (Base CALs) for each accessing user or device or, alternatively, a Windows Server External Connector license (Base EC) for each server that is accessed. External access to advanced functionality on licensed servers requires the CAL associated with that functionality (Additive CAL) for each accessing user or device or, alternatively, the External Connector license associated with that functionality (Additive EC) for each server accessed. Additive External Connector Licenses must be assigned in addition to the corresponding Base access license(s) for access to advanced functionality on the licensed server.

**Access provided:** Windows Server 2025 CALs and External Connectors licenses permit access to Windows Server 2025 server software and prior versions of Windows Server software. To learn more about server access licenses, see the Microsoft server Base and Additive Client Access Licenses Overview Licensing brief.

Each user CAL permits one user, using any device, to access instances of the server software on their licensed servers. Each device CAL permits one device, used by any user, to access instances of the server software on their licensed servers. External Connector licenses permit external user access to instances of the server software on the licensed servers. Customers may use a combination of User and Device CALs, and External Connector licenses as required. To learn more about indirect-access licensing, see the Multiplexing - Overview Licensing brief.

# Requirements for internal users – Client Access License (CAL)

For users that are either the customer's or its affiliates' employees, or its or its affiliates' onsite contractors or onsite agents:

| Access License | Category | Datacenter | Standard | Essentials |
|---|---|---|---|---|
| Windows Server | Base | CAL [1] | CAL [1] | n/a |
| Windows Server Remote Desktop Services | Additive | CAL [2],[3] | CAL [2],[3] | n/a [4] |
| Windows Server Active Directory Rights Management Services | Additive | CAL [2] | CAL [2] | CAL [5] |
| Microsoft Identity Manager | Additive | CAL [2],[6] | CAL [2],[6] | n/a |

**Notes:**

[1] Required for every user or device accessing the licensed server.

[2] Required (in addition to Base CAL) for every user or device accessing the advanced functionality.

[3] Also required (in addition to Base CAL) for use of Windows Server to host a graphical user interface.

[4] Not required for Remote Web Access feature of the Essentials edition. Only the RD Gateway role service is installed and configured, other RDS role services including RD Session Host are not supported.

[5] Required for each User Account through which a user directly or indirectly accesses the advanced functionality.

[6] Also required (in addition to Base CAL) for any person for whom the software issues or manages identity information.

# Requirements for external users – Client Access License (CAL) and/or External Connector license (EC)

For users that are not either customer's or its affiliates' employees, or its or its affiliates' onsite contractors or onsite agents:

| Access License | Category | Datacenter | Standard |
|---|---|---|---|
| Windows Server | Base | CAL [1] / EC [7] | CAL [1] / EC [7] |
| Windows Server Remote Desktop Services | Additive | CAL [2],[3] / EC [8], [3] | CAL [2],[3] / EC [8], [3] |
| Windows Server Active Directory Rights Management Services | Additive | CAL [2] / EC [8] | CAL [2] / EC [8] |
| Microsoft Identity Manager | Additive | CAL [2],[6] / EC [8], [9] | CAL [2],[6] / EC [8], [9] |

**Notes:**

[1] Required for every user or device accessing the licensed server.

[2] Required for every user or device accessing the licensed server, if the advanced functionality is accessed.

[3] Also required for use of Windows Server to host a graphical user interface.

[4] Not required for Remote Web Access feature of the Essentials edition. Only the RD Gateway role service is installed and configured, other RDS role services including RD Session Host are not supported.

[5] Required for each user account through which a user directly or indirectly accesses the advanced functionality.

[6] Also required for any person for whom the software issues or manages identity information.

[7] Required for each physical server (any number of OSEs) that is accessed.

[8] Required for each physical server (any number of OSEs) that is accessed, if the advanced functionality is accessed.

[9] Also required for any external user for whom the software issues or manages identity information (in absence of CALs).

# Use of Client Access Licenses (CALs) or External Connectors (ECs)

CALs and ECs permit access to instances of the server software on licensed servers. CALs and ECs permit access to the same version, or earlier versions, of the server software. All server software access requires CALs, ECs, or CAL Equivalent licenses. CAL equivalent licenses include User Subscription Licenses (SLs) corresponding to the base or advanced functionality. Examples include the Core CAL, the Microsoft 365 E3/E5 User SL, and individual service SLs, such as Azure Information Protection P1 User SL and Azure Active Directory Premium User SL. For more information about which CAL equivalent licenses correspond to Windows Server, see the Product Terms.

| | **Windows Server 2025** | **Windows Server 2022** | **Windows Server 2019** |
|---|---|---|---|
| 2025 CAL/EC | Yes | Yes | Yes |
| 2022 CAL/EC | No | Yes | Yes |
| 2019 CAL/EC | No | No | Yes |

CALs and ECs are not required:

- For access by another licensed server (for example, one licensed server accessing another licensed server).
- To access server software running a web workload (such as content served within an Internet web solution on a publicly available website) or high-performance computing (HPC) workload (such as server software used to run a cluster node, in conjunction with other software on a cluster node, for the purposes of supporting the clustered HPC applications).
- For access in a physical OSE used solely for hosting and managing virtual OSEs (for example, if 2022 is used in a physical OSE as the hypervisor, but all virtual OSEs are 2019, only 2019 CALs or ECs are required).

# Software Assurance benefits

Microsoft Commercial Licensing customers with active Software Assurance get more out of their Microsoft software and services investments with access to a unique set of technologies, services, and license rights to help use Microsoft products efficiently. Here is a list of just a few of the benefits that customers can receive with Software Assurance for Windows Server:

**New version rights:** Upgrade each product license covered by active Software Assurance to the most recent version when available.

**Step-up licenses:** Customers with active Software Assurance can migrate from a lower- to higher-level edition of certain products (for example, from Windows Server 2025 Standard to Windows Server 2025 Datacenter). Note that the Step-up license option is not available through the Open License.

**Back-up for disaster recovery:** Provides additional instances for servers used as offline (“cold”) backups, to help customers recover in case of a catastrophic event. See below for additional details.

**Azure Hybrid Benefit for Windows Server:** Customer may upload and use their own Windows Server images on Microsoft Azure through Azure Virtual Machines (“Base Instances”) and pay only for the cost of service utilization of the Base Instances. To learn more see here.

**Flexible Virtualization Benefit:** Expands your outsourcing options for software to include Authorized Outsourcers’ shared servers. This benefit is available broadly for software products under your subscription licenses and licenses with Software Assurance. For more information, see the Flexible Virtualization Benefit Licensing guide.

# Disaster recovery rights

For each instance of eligible server software that a customer runs in a physical or virtual OSE on a licensed server, they may temporarily run a backup instance in a physical or virtual OSE on either (a) servers dedicated to disaster recovery and to the customer's use, or (b) for instances of eligible software other than Windows Server, on Microsoft Azure Services, provided that the backup instance is managed by Azure Site Recovery to Azure. The license terms for the software and the following limitations apply to the customer's use of the backup instance.

The backup instance can run only during the following exception periods:

- For brief periods of disaster recovery testing within one week every 90 days.
- During a disaster, while the production server being recovered is down.
- Around the time of a disaster, for a brief period, to assist in the transfer between the primary production server and the disaster recovery server.

To use the software under disaster recovery rights, the customer must comply with the following terms:

- The OSE on the disaster recovery server must not be running at any other times except as above.
- The OSE on the disaster recovery server may not be in the same cluster as the production server.
- Other than backup instances running on Microsoft Azure Services, a Windows Server license is not required for the disaster recovery server if the Hyper-V role within Windows Server is used to replicate virtual OSEs from the production server at a primary site to a disaster recovery server.
- The disaster recovery server may be used only to run hardware virtualization software (such as Hyper-V), provide hardware virtualization services, run software agents to manage the hardware virtualization software, serve as a destination for replication, receive replicated virtual OSEs, test failover, and/or await failover of the virtual OSEs.
- Run disaster recovery workloads as described above.
- The disaster recovery server may not be used as a production server.
- Use of the software backup instance should comply with the license terms for the software.
- Once the disaster recovery process is complete and the production server is recovered, the backup instance must not be running at any other times except those times allowed here.
- Maintain Software Assurance coverage for all CALs, External Connector licenses, and Server Management Licenses to access the backup instance, gain access to servers with active ESU coverage, and manage the OSEs where the software runs.
- The customer's right to run the backup instances ends when their Software Assurance coverage ends.

Specific Software Assurance benefit eligibility can vary by Commercial Licensing program (such as an Enterprise Agreement or Open Value Agreement) and product, as well as the number of qualifying licenses that are enrolled in Software Assurance. Visit Software Assurance for more information.

# Original equipment manufacturer (OEM) licenses – enrollment into Software Assurance

Software Assurance is available for the latest version (determined by the most recent version available on the Commercial Licensing Service Center) of the software: Windows Server 2025. Customers must acquire Software Assurance within 90 days of purchase (see the Product Terms, Software Assurance section for details). If a customer acquires Software Assurance for an OEM license, their use of the software becomes subject to the Microsoft Product Terms for that product and the terms and conditions or their organization's Commercial Licensing agreement.

# How to buy Windows Server licenses

Windows Server software licenses are sold through channels designed to meet the unique needs of customers. These sales channels include online retailers offering full packaged product (FPP) licenses of Windows Server software, original equipment manufacturers (OEMs) offering pre-installed licenses with their hardware systems, as well as Licensing Solutions Partners (LSPs) and Enterprise Software Advisors (ESAs) offering Windows Server software through Microsoft Commercial Licensing programs for end-customer organizations.

## Commercial Licensing

For organizations with as few as five users, Microsoft offers licensing programs to help reduce administrative overhead and software management costs, while enabling product licensing on an ongoing basis at a considerable discount. The various licensing options enable customers to choose the program that works best for their management and operational needs.

Comprehensive programs that offer Software Assurance as a fixed benefit: Open Value (OV), Open Value Subscription (OVS), Enterprise Agreement, Enterprise Agreement subscription, Server and Cloud Enrollment (SCE).

Transactional programs: Microsoft Customer Agreement and the Microsoft Products and Services Agreement (MPSA). Windows Server is not available through the Microsoft Independent Software Vendor (ISV) Royalty Licensing Program.

Microsoft also offers programs that can meet the specific needs of organizations that partner with Microsoft to provide additional software and services, such as the Microsoft Services Provider License Agreement (SPLA).

## Server and Cloud Enrollment (SCE)

The Server and Cloud Enrollment is an enrollment under the Microsoft Enterprise Agreement that enables highly committed customers to standardize broadly on one or more key server and cloud technologies from Microsoft. In exchange for making an installed base-wide commitment to one or more components of the Server and Cloud Enrollment, customers receive the best pricing and terms, plus other benefits, including cloud-optimized licensing options and simplified license management.

## Windows Server pay-as-you-go enabled by Azure Arc

A new cloud billing option for Windows Server customers, providing pay-as-you-go flexibility across on-premises and cloud environments.

Pay as you go allows customers without unlimited virtualization the option to scale up with additional virtual machines to meet capacity bursts or temporary needs, or to deploy more servers as needed. Customers can connect to Azure Arc and transact directly in the portal.

To learn more about how to use pay-as-you-go for Windows Server 2025, please see Configure Windows Server Pay-as-you-go with Azure Arc | Microsoft Learn and Windows Server 2025 Licensing & Pricing | Microsoft.

## Channel availability

This table below shows the primary channel availability for Windows Server 2025 software licenses. Every edition may not be available in all channels or licensing programs in all regions. For details about Microsoft Licensing Programs, visit the Licensing Resources and Documents site, Programs page.

| Windows Server 2025 | EA/EAS/SCE | MPSA | MCA | SPLA | OV/OVS | FPP/ESD | OEM |
|---|---|---|---|---|---|---|---|
| Datacenter edition | ü | ü | ü | ü | | ü | ü |
| Standard edition | ü | ü | ü | ü | | ü | ü |
| Essentials edition | | | | | | | ü |

**Notes:**

*EA/EAS/SCE = Enterprise Agreement, Enterprise Subscription Agreement, and Server and Cloud Enrollment*

*MPSA = Microsoft Products and Services Agreement*

*MCA = Microsoft Customer Agreement*

*SPLA = Services Provider License Agreement*

*OV/OVS = Open Value/Open Value Subscription Licensing Program*

*FPP/ESD = Full Package Product and Electronic Software Download*

*OEM = original equipment manufacturer*

# Availability through Commercial Licensing programs

Licenses for Windows Server editions are available in the following Microsoft Commercial Licensing programs.

| | OL [1] | S/S+ [2] | MPSA [3] | OV/OVS [4] | EA/EAS [5] | OVS-ES [6] | EES [7] | MCA [8] |
|---|---|---|---|---|---|---|---|---|
| Datacenter Edition | ü | ü | ü | ü | ü | ü | ü | ü |
| Standard Edition | ü | ü | ü | ü | ü | ü | ü | ü |

**Notes:**

*[1] OL = Open License (The Open License program was retired December 31, 2021. New licenses are no longer available as of January 1, 2022.*

*[2] S/S+ = Select and Select Plus*

*[3] MPSA = Microsoft Products and Services Agreement*

*[4] OV/OVS = Open Value and Open Value Subscription*

*[5] EA/EAS = Enterprise Agreement and Enterprise Subscription Agreement*

*[6] OVS-ES = Open Value Subscription – Education Solutions*

*[7] EES = Enrollment for Education Solutions*

*[8] MCA = Microsoft Customer Agreement*

# Volume Activation

Volume Activation is a set of technologies and tools designed to automate the product activation process for systems that are deployed under a Microsoft Commercial Licensing agreement. Windows Server 2025 includes technologies designed to simplify the task of configuring the distribution and management of an organization's volume software licenses. For more information, visit Volume Activation Overview.

**The Volume Activation Services server role:** Volume Activation Services is a server role in Windows Server (2012 or later editions) that enables a customer to automate and simplify the issuance and management of Microsoft software volume licenses for a variety of scenarios and environments. With Volume Activation Services, customers can install and configure the Key Management Service (KMS) and enable Active Directory-based Activation.

**Key Management Service (KMS):** KMS is a role service that allows organizations to activate systems within their network from a server where a KMS host has been installed. With KMS, IT pros can complete activations on their local network, eliminating the need for individual computers to connect to Microsoft for product activation. KMS does not require a dedicated system, and it can be cohosted on a system that provides other services. By default, volume editions of Windows client and server operating systems connect to a system that hosts the KMS service to request activation. No action is required from the user.

**Active Directory-based activation:** Active Directory-based activation is a role service that allows the customer to use Active Directory Domain Services (Active Directory DS) to store activation objects, which can further simplify the task of maintaining volume activation services for a network. With Active Directory-based activation, no additional host server is needed, and activation requests are processed during computer startup.

Any computers running Windows Server 2016 (or later editions) with a Generic Volume License Key (GVLK) that are connected to the domain will activate automatically and transparently. They will stay activated as long as they remain members of the domain and maintain periodic contact with a domain controller. Activation takes place after the licensing service starts. When this service starts, the computer running Windows Server 2016 (or later editions) contacts Active Directory DS automatically, receives the activation object, and activates without user intervention.

# Downgrade rights

Windows Server is streamlined and simple, making it easy for customers to choose the edition that is right for their needs. Choose from three primary editions of Windows Server, based on organization size as well as virtualization and datacenter requirements. Customers can also choose to deploy earlier

versions of the same edition or lower editions in place of what they are licensed for. These are referred to as “downgrade rights” and “down edition rights.” When invoking downgrade rights or down edition rights, the license terms of the version and edition acquired still govern use of the software.

| Licensed edition | Down editions | Versions [1] | Use rights [2] |
|---|---|---|---|
| Windows Server 2025 Datacenter | Windows Server Datacenter | 2022 or earlier | 2025 |
| | Windows Server Standard | 2022 or earlier | 2025 |
| | Windows Server Essentials | 2022 or earlier | 2025 |
| Windows Server 2025 Standard | Windows Server Standard | 2022 or earlier | 2025 |
| | Windows Server Essentials | 2022 or earlier | 2025 |

**Notes:**

*[1] Customers may downgrade to any other version as long as they have the appropriate media and keys.*

*[2] All use rights are still governed by the licensed edition.*

# Planning for Windows Server 2025

When planning to deploy Windows Server 2025, either through upgrades or new licenses, remember the following:

Select the edition of Windows Server 2025 based on virtualization needs and edition features:

**Datacenter** edition for highly virtualized and software-defined datacenter environments

**Standard** edition for low-density or non-virtualized environments

Renewing Software Assurance is the best way to protect investments and provide access to new versions, Deployment Planning Services, and technical assistance

Core CAL and Enterprise CAL Suites will continue to be the most cost-effective way to purchase Windows Server CALs to access workloads running on Windows Server 2025 Standard and Datacenter editions.

The Microsoft Core Infrastructure Suite (CIS) will continue to offer the best value for private cloud and datacenter management pricing.

Prices and pricing levels vary. Microsoft does not determine pricing or payment terms for licenses acquired through resellers. Microsoft reaches customers at sales offices, support centers, and technology centers around the world. For specific pricing, connect with a Microsoft reseller at a sales office.

# FAQs

See more FAQs.

# Licensing definitions

**Authorized Outsourcer** means any third party service provider that is not a Listed Provider and is not using Listed Provider as a Data Center Provider as part of the outsourcing service.

**Client Access License (CAL)** means client access license, which may be assigned by user or device, as appropriate. A user CAL allows access to corresponding version of the server software or earlier versions of the server software from any device by one user. A device CAL allows access to corresponding versions of the server software or earlier versions of the server software from one device by any user. CALs allow access to server software running on a customer's licensed servers only.

**Clustered HPC application** means a high-performance computing application that solves, in parallel, complex computational problems, or a set of closely related computational problems. clustered HPC applications divide a computationally complex problem into a set of jobs and tasks which are coordinated by a job scheduler, such as provided by Microsoft HPC Pack, or similar HPC middleware, which distributes these in parallel across one or more computers operating within an HPC cluster.

**Cluster node** means a device that is dedicated to running clustered HPC applications or providing job scheduling services for clustered HPC applications.

**External Connector** licenses means a license assigned to a server dedicated to customer's use that permits access to the corresponding version of the server software or earlier versions of the server software by external users.

**External users** means users that are not (a) employees of Customer or its Affiliates, (b) contractors or agents that typically work for Customer or its Affiliates for more than 30 hours on average per week, or (c) contractors or agents that typically work onsite for Customer or its Affiliates on each working day.

**High-performance computing (HPC) workload** means a workload where the server software is used to run a cluster node and is used in conjunction with other software as necessary to permit security, storage, performance enhancement and systems management on a cluster node for the purpose of supporting the clustered HPC applications.

**Hyper-V container** is a feature of Windows Server that utilizes a virtual operating system environment. Each Hyper-V container is considered to be one virtual OSE.

**Licensed server** means a single server, dedicated to customer use, to which a license is assigned. For purposes of this definition, a hardware partition or blade is considered to be a separate server.

**License** means the right to download, access, install, and use a product.

**Listed Providers** include entities identified by Microsoft at http://aka.ms/listedproviders. Microsoft may identify additional Listed Providers at http://aka.ms/listedproviders from time to time; however, if Customer is using an outsourcer at the time its Authorized Outsourcer status is terminated, then Customer may temporarily continue to use the same entity in its former Authorized Outsourcer capacity for one year from the date of that change in status.

**Operating system environment (OSE)** means all or part of an operating system instance, or all or part of a virtual (or otherwise emulated) operating system instance which enables separate machine identity (primary computer name or similar unique identifier) or separate administrative rights, and instances of applications, if any, configured to run on the operating system instance or parts identified above. A physical hardware system can have one physical OSE and/or one or more virtual OSEs.

**Physical core** means a core in a physical processor.

**Physical OSE** means an OSE that is configured to run directly on a physical hardware system. The operating system instance used to run hardware virtualization software or to provide hardware virtualization services is considered part of the physical OSE.

**Physical processor** means a processor in a physical hardware system.

**Server** means a physical hardware system capable of running server software.

**Virtual OSE** means an OSE that is configured to run on a virtual hardware system.

**Web workload** (also referred to as "Internet web solutions") are publicly available web pages, websites, web applications, web services, and/or POP3 mail serving. For clarity, access to content, information, and applications served by the software within an Internet web solution is not limited to a customer's or its affiliates' employees. Software in Internet web solutions is used to run:

- Web server software (for example, Microsoft Internet Information Services), and management or security agents (for example, the System Center Operations Manager agent);
- Database engine software (for example, Microsoft SQL Server) solely to support Internet web solutions; or
- The Domain Name System (DNS) service to provide resolution of Internet names to IP addresses as long as that is not the sole function of that instance of the software.

**Windows Server container** is a feature of Windows Server software.

# For more information

**Windows Server Licensing Resources:** https://www.microsoft.com/licensing/docs/view/Windows-Server

**Microsoft Licensing Resources and Documents:** https://www.microsoft.com/licensing/docs

**Microsoft Commercial Licensing:** www.microsoft.com/licensing

**Microsoft M365 admin center:** https://admin.microsoft.com

**Activation:** https://learn.microsoft.com/en-us/windows-server/get-started/kms-client-activation-keys

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Copilot for organizations
Copilot for personal use
AI in Windows
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Windows 11 apps
Download Center
Microsoft Store support
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Order tracking
Certified Refurbished
Microsoft Store Promise
Flexible Payments
Devices for education
Microsoft Teams for Education
Microsoft 365 Education
How to buy for your school
Educator training and development
Deals for students and parents
AI for education
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Microsoft Power Platform
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